Thomas Cohn

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Education

Massachusetts Institute of Technology *Cambridge, Massachusetts*

Ph.D. in Computer Science and Engineering *Sep. 2022 - Present*

- Advisor: Russ Tedrake
- GPA 5.00/5.00

S.M. in Electrical Engineering and Computer Science *Sep. 2022 - May 2024*

- Advisor: Russ Tedrake
- Thesis Title: Motion Planning along Manifolds with Geodesic Convexity and Analytic Inverse Kinematics

University of Michigan *Ann Arbor, Michigan*

B.S.E. Computer Science and Engineering *Sep. 2017 - May 2022*
B.S. Honors Mathematics

- Magna Cum Laude
- Engineering Honors Program
- Minors in Statistics and Music
- GPA 3.74/4.00

Honors and Awards

Selected Honors and Awards

2024 **National Science Foundation Graduate Research Fellowship**
2024 **Best Paper Award in Robot Manipulation Finalist,** ICRA
2023 **Best Paper Award Finalist,** RSS
2022 **Outstanding Undergraduate Research Award,** University of Michigan
2021 **1st Place Award,** University of Michigan Engineering Research Symposium

Other Honors and Awards

2026 **Best Contribution Runner Up,** RSS Workshop on the Geometry of Motion
2026 **Best Workshop Paper Award,** RSS Workshop on Data-Centric Robotics
2026 **Most Useful Practical Information Award,** RSS Workshop “It’s the Demos”
2025 **Best Poster Award,** IROS Workshop on Robotic Data Generation and Evaluation (RoDGE)
2025 **Best Workshop Paper Award,** IROS Workshop on Frontiers in Dynamic, Intelligent, and Adaptive Multi-Arm Manipulation
2025 **2nd Place,** MIT-Amazon Robotics Graduate Student Research Presentation Competition

Publications

*Denotes equal contribution.

Conference Publications

[C5] Nicholas Pfaff, **Thomas Cohn**, Sergey Zakharov, Rick Cory, and Russ Tedrake. “SceneSmith: Agentic Generation of Simulation-Ready Indoor Scenes”. In: *Forty-third International Conference on Machine Learning*. 2026. **Spotlight**.

[C4] **Thomas Cohn** and Russ Tedrake. “Sampling-Based Motion Planning with Discrete Configuration-Space Symmetries”. In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2025, pp. 13119–13126.

[C3] **Thomas Cohn**, Seiji Shaw, Max Simchowitz, and Russ Tedrake. “Constrained bimanual planning with analytic inverse kinematics”. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2024, pp. 6935–6942. **Best Paper in Robot Manipulation Award Finalist**.

[C2] **Thomas Cohn**, Mark Petersen, Max Simchowitz, and Russ Tedrake. “Non-Euclidean Motion Planning with Graphs of Geodesically-Convex Sets”. In: *Proceedings of Robotics: Science and Systems (RSS)*. Daegu, Republic of Korea, July 2023. **Best Paper Award Finalist**.

[C1] **Thomas Cohn**, Nikhil Devraj, and Odest Chadwicke Jenkins. “Topologically-informed atlas learning”. In: *2022 International Conference on Robotics and Automation (ICRA)*. IEEE. 2022, pp. 3598–3604.

Journal Publications

- [J4] Peter Werner*, **Thomas Cohn***, Rebecca H. Jiang, Tim Seyde, Max Simchowitz, Russ Tedrake, and Daniela Rus. "Faster algorithms for growing collision-free convex polytopes in robot configuration space". In: *The International Journal of Robotics Research* (2026).
- [J3] **Thomas Cohn**, Mark Petersen, Max Simchowitz, and Russ Tedrake. "Non-Euclidean motion planning with graphs of geodesically convex sets". In: *The International Journal of Robotics Research* 44.10-11 (2025), pp. 1840–1862.
- [J2] Shruti Garg, **Thomas Cohn**, and Russ Tedrake. "Planning Shorter Paths in Graphs of Convex Sets by Undistorting Parametrized Configuration Spaces". In: *IEEE Robotics and Automation Letters* (2025).
- [J1] **Thomas Cohn**, Odest Chadwicke Jenkins, Karthik Desingh, and Zhen Zeng. "TSBP: Tangent Space Belief Propagation for Manifold Learning". In: *IEEE Robotics and Automation Letters* 5.4 (2020), pp. 6694–6701.

Preprints

- [P5] **Thomas Cohn***, Seiji Shaw*, Harel Biggie, Travis Manderson, Nicholas Roy, and Russ Tedrake. "Planning along Differentiable Charts of Constraint Manifolds with General-Purpose IK Solvers". In: *arXiv preprint arXiv:2609.10905* (2026).
- [P4] Adam Wei, Nicholas Pfaff*, **Thomas Cohn***, Arif Kerem Dayı, Constantinos Daskalakis, Giannis Daras, and Russ Tedrake. "Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics". In: *arXiv preprint arXiv:2606.12365* (2026). **Accepted for Publication at CoRL 2026.**
- [P3] **Thomas Cohn***, Lihan Tang*, Alexandre Amice, and Russ Tedrake. "A Framework for Combining Optimization-Based and Analytic Inverse Kinematics". In: *arXiv preprint arXiv:2602.05092* (2026).
- [P2] Lexi Foland, **Thomas Cohn**, Adam Wei, Nicholas Pfaff, Boyuan Chen, and Russ Tedrake. "How Well do Diffusion Policies Learn Kinematic Constraint Manifolds?" In: *arXiv preprint arXiv:2510.01404* (2025). **Accepted for Publication at ICRA 2026.**
- [P1] Peter Werner, **Thomas Cohn***, Rebecca H. Jiang*, Tim Seyde, Max Simchowitz, Russ Tedrake, and Daniela Rus. "Faster Algorithms for Growing Collision-Free Convex Polytopes in Robot Configuration Space". In: *arXiv preprint arXiv:2410.12649* (2024). **Accepted for Publication at ISRR 2024.**

Workshop Papers

- [W7] **Thomas Cohn**, Seiji Shaw, Nicholas Roy, and Russ Tedrake. "Principled Domain Extension for Analytic IK". In: *The Geometry of Motion: Physics-Informed Structures for Learning and Control*. *Robotics: Science and Systems (RSS)*. 2026. **Best Contribution Runner Up**.
- [W6] Adam Wei, Nicholas Pfaff*, **Thomas Cohn***, Arif Kerem Dayı, Constantinos Daskalakis, Giannis Daras, and Russ Tedrake. "Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics". In: *Data-Centric Robotics: What Data Do Robots Really Need? Robotics: Science and Systems (RSS)*. 2026. **Spotlight, Best Workshop Paper Award**.
- [W5] Adam Wei, Nicholas Pfaff*, **Thomas Cohn***, Arif Kerem Dayı, Constantinos Daskalakis, Giannis Daras, and Russ Tedrake. "Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics". In: *It's the demos: A Deep Look at the Role of Demonstration Quality in Imitation-Based Robot Manipulation*. *Robotics: Science and Systems (RSS)*. 2026. **Spotlight, Most Useful Practical Information Award**.
- [W4] **Thomas Cohn**, Seiji Shaw, Nicholas Roy, and Russ Tedrake. "Planning along Differentiable Charts of Constraint Manifolds with the Inverse Function Theorem". In: *Geometry in the Age of Data-Driven Robotics*. *2026 International Conference on Robotics and Automation (ICRA)*. 2026.
- [W3] **Thomas Cohn***, Lihan Tang*, Alexandre Amice, and Russ Tedrake. "A Framework for Combining Optimization-Based and Analytic Inverse Kinematics". In: *2nd Workshop on Frontiers of Optimization for Robotics*. *2026 International Conference on Robotics and Automation (ICRA)*. 2026.
- [W2] Lexi Foland, **Thomas Cohn**, Adam Wei, Nicholas Pfaff, Boyuan Chen, and Russ Tedrake. "How Well do Diffusion Policies Learn Kinematic Constraint Manifolds?" In: *RoDGE: Robotic Data Generation and Evaluation*. *2025 International Conference on Intelligent Robots and Systems (IROS)*. 2025. **Best Poster Award**.
- [W1] **Thomas Cohn**, Peter Werner, and Russ Tedrake. "Faster Algorithms for Growing Collision-Free Regions of Constrained Bimanual Configuration Spaces". In: *Frontiers in Dynamic, Intelligent, and Adaptive Multi-Arm Manipulation*. *2025 International Conference on Intelligent Robots and Systems (IROS)*. 2025. **Best Workshop Paper Award**.

Presentations

Invited Talks

2026 **Invited Talk,** "Motion Planning in Minimal Coordinates for Kinematically-Constrained Systems"
RSS Workshop on the Geometry of Motion – (Recording)

2025 **Invited Talk,** "Non-Euclidean Motion Planning with Graphs of Geodesically-Convex Sets"
ICCOPT

Conference Presentations

2025 **Oral + Poster Presentation,** "Sampling-Based Motion Planning with Discrete Configuration-Space Symmetries"
IROS

2024 **Oral + Poster Presentation,** "Constrained Bimanual Planning with Analytic Inverse Kinematics"
ICRA

2023 **Oral + Poster Presentation,** "Non-Euclidean Motion Planning with Graphs of Geodesically-Convex Sets"
RSS

2022 **Oral + Poster Presentation,** "Topologically-Informed Atlas Learning"
ICRA

2020 **Oral Presentation,** "TSBP: Tangent Space Belief Propagation for Manifold Learning"
IROS

Workshop Presentations

2026 **Poster Presentation,** "Principled Domain Extension for Analytic IK"
RSS Workshop on the Geometry of Motion

2026 **Poster Presentation,** "Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics"
RSS Workshop on Data-Centric Robotics

2026 **Poster Presentation,** "Ambient Diffusion Policy: Imitation Learning from Suboptimal Data in Robotics"
RSS Workshop on It's the demos

2026 **Poster Presentation,** "Planning along Differentiable Charts of Constraint Manifolds with the Inverse Function Theorem"
ICRA Workshop on Geometry in the Age of Data-Driven Robotics

2026 **Poster Presentation,** "A Framework for Combining Optimization-Based and Analytic Inverse Kinematics"
ICRA Workshop on Frontiers of Optimization for Robotics

2025 **Poster Presentation,** "How Well do Diffusion Policies Learn Kinematic Constraint Manifolds?"
IROS Workshop on RoDGE: Robotic Data Generation and Evaluation

2025 **Spotlight Talk + Poster Presentation,** "Faster Algorithms for Growing Collision-Free Regions of Constrained Bimanual Configuration Spaces"
IROS Workshop on Frontiers in Dynamic, Intelligent, and Adaptive Multi-Arm Manipulation

Other Presentations

2025 **Oral Presentation,** "Non-Euclidean Motion Planning with Graphs of Geodesically-Convex Sets"
LIDS Student Conference

2023 **Poster Presentation,** "Constrained Bimanual Planning with Analytic Inverse Kinematics"
Northeast Robotics Colloquium

2021 **Poster Presentation,** "Topologically-Informed Atlas Learning"
University of Michigan Engineering Research Symposium – **1st Place Award**

2021 **Poster Presentation,** "Coordinate Chart Particle Filter for Deformable Object Pose Estimation"
University of Michigan Engineering Research Symposium

2019 **Poster Presentation,** "TSBP: Tangent Space Belief Propagation for Manifold Learning"
University of Michigan Engeineering Research Symposium

Grants and Fellowships

2024 **Graduate Research Fellowship,** National Science Foundation
2022 **Frederick and Barbara Cronin Fellowship,** Massachuetts Institute of Technology
2020 **Raab Family Scholarship,** University of Michigan Marching Band
2019 **Wanda W. Lincoln Scholarship,** University of Michigan Marching Band
2017 **The Gloria Wille Bell and Carlos R. Bell Scholarship**
2017 **Regents Merit Scholarship,** University of Michigan

Students Mentored (Research Supervision)

Students who co-authored publications or preprints are indicated with *.

Current Students

2026- **Rinoa Oliver,** MIT Undergraduate Student.
2025- **Lexi Foland*,** MIT Undergraduate Student.
2025- **Lihan Tang*,** MIT Undergraduate Student.
2025- **Margulan Ismoldayev,** MIT Undergraduate Student.

Former Students

2023-2025 **Shruti Garg*,** MIT Undergraduate and Masters Student. *Starting PhD at University of Pennsylvania in Fall 2026.*

Teaching

2024 (Spring)	**CSCI 5551: Introduction to Intelligent Robotic Systems,** Guest Lecture *Faculty Instructor: Karthik Desingh*	*University of Minnesota*
2023 (Fall)	**6.4210: Robotic Manipulation,** Teaching Assistant *Faculty Instructor: Russ Tedrake*	*Massachusetts Institute of Technology*
2022 (Winter)	**EECS 367: Introduction to Autonomous Robotics,** Teaching Assistant *Faculty Instructor: Chad Jenkins*	*University of Michigan*
2021 (Fall)	**ROB 102: Introduction to AI and Programming,** Teaching Assistant *Faculty Instructor: Chad Jenkins*	*University of Michigan*
2020 (Winter)	**ENGR 100-250: Microprocessors and Toys,** Teaching Assistant *Faculty Instructor: Peter Chen*	*University of Michigan*
2019 (Winter)	**ENGR 100-250: Microprocessors and Toys,** Teaching Assistant *Faculty Instructor: Peter Chen*	*University of Michigan*

Work Experience

2022-	**Graduate Student Research Assistant,** Massachusetts Institute of Technology, PI: Russ Tedrake
2016-2022	**Undergraduate Student Research Assistant,** University of Michigan, PI: Chad Jenkins
2021	**Curriculum Designer,** Robotics @ Marygrove
2017-2018	**Software Developer,** Number DNA

Extracurriculars

2022-Present	**MIT Graduate Hillel,** GradHillel President 2024-2025, Board of Directors 2024-Present
2017-2022	**Michigan Marching Band,** Cymbal Section Leader 2019-2022
2017-2022	**Michigan Hockey Pep Band**
2018-2020	**Michigan Percussion Chamber Ensemble**

Service

2024-Present	**Reviewer,** ICRA, RA-L, IROS, Acta Astronautica, Humanoids, T-RO, L-CSS, T-MECH, Neurocomputing, T-IE, RSS, T-ASE, IJRR
2024-Present	**Graduate School Application Mentor,** MIT GAAP Program